

LORMA COLLEGES
COLLEGE OF COMPUTER STUDIES AND ENGINEERING
Urbiztondo, San Juan, La Union, Philippines 2514

PARKING ASSIST SYSTEM

RANDY JORDAN PINEDA MACABITAS JR.
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P411 | EMBEDDED SYSTEM

Under the instruction of
Engr. BRIANNE MARK AQUINO

INTRODUCTION

Parking vehicles safely can sometimes be difficult, especially in tight parking spaces or areas with limited visibility. Drivers may accidentally hit walls, posts, or other vehicles because they cannot clearly estimate the remaining distance between the car and nearby objects. Because of this, modern vehicles are now commonly equipped with parking assist technologies that help drivers park more safely and accurately.

A Parking Assist System is an electronic safety feature that uses sensors to detect nearby objects and provide warnings to the driver. These systems usually measure the distance between the vehicle and obstacles, then notify the driver through visual displays and sound alerts. According to automotive safety studies, parking sensors significantly reduce minor collisions that occur during parking maneuvers.

This project demonstrates a simplified version of a real car parking assist system using embedded system technology. The prototype uses an ultrasonic sensor to measure the distance between the vehicle and an object. The Arduino microcontroller processes the measured distance and displays the information on an LCD screen while activating a buzzer that changes its beep speed depending on how close the object is.

When the object is far away, the buzzer remains silent and the LCD shows that the distance is safe. As the object gets closer, the buzzer starts to beep slowly and then faster to warn the driver. When the distance becomes very small, the system activates a continuous alarm and displays a stop warning.

This project shows how embedded systems can be used to improve vehicle safety by providing drivers with real-time feedback during parking.

OBJECTIVES

- To design a simple parking assistance prototype using an ultrasonic distance sensor.

- To measure the distance between the vehicle and nearby objects in real time.
- To display the measured distance using a 16x2 LCD screen.
- To provide audible warnings using a buzzer that changes beep speed depending on the distance detected.
- To demonstrate how embedded systems can be applied in automotive safety technologies.

SYSTEM BLOCK DIAGRAM (IPO)

The Parking Assist System follows the Input–Process–Output (IPO) model. The system collects distance information through a sensor, processes the data using a microcontroller, and provides visual and audio warnings as output.

INPUT

- Ultrasonic Sensor (Distance Detection)

The ultrasonic sensor continuously sends ultrasonic waves and receives the reflected signal to determine the distance between the system and an object.

PROCESS

- Arduino Uno Microcontroller

The Arduino processes the sensor data and compares the measured distance to predefined threshold values:

- Distance greater than 40 cm → Safe distance
- Distance between 20–40 cm → Getting close
- Distance between 10–20 cm → Very close
- Distance less than 10 cm → Stop warning

The Arduino also controls the LCD display and buzzer behavior based on these conditions.

OUTPUT

- LCD Display – Shows the parking status and distance measurement.
- Buzzer – Produces warning sounds that increase in frequency as the object gets closer.

LIST OF COMPONENTS

The Parking Assist System uses several electronic components to detect distance and provide warnings to the user. The main controller of the system is the Arduino Uno, which processes the sensor data and controls the outputs.

An ultrasonic sensor is used to measure the distance between the system and nearby objects. A 16x2 LCD screen is used to display the distance and system messages in real time. An active buzzer provides audio alerts when objects are detected at close distances.

The system also uses a breadboard and jumper wires to connect all the electronic components together

COMPONENT	QUANTITY	PURPOSE
Arduino Uno	1	Main controller
Ultrasonic Sensor	1	Measures distance to object
LCD	1	Displays distance and warnings
Active buzzer	1	Provides audible alerts
Breadboard	1	Circuit prototyping platform
Jumper Wires	Several	Electrical connections
USB cable	1	Powers the system

SCHEMATIC DIAGRAM

The Parking Assist System is centered around the Arduino Uno microcontroller, which processes distance measurements and controls the output devices.

The ultrasonic sensor is connected to the Arduino and is responsible for measuring the distance between the system and nearby obstacles. The sensor sends ultrasonic waves and calculates the distance based on the time taken for the echo signal to return.

The LCD display is connected to the Arduino to show the parking status and the measured distance. The buzzer is also connected to a digital pin of the Arduino and produces warning sounds when the detected object becomes closer.

All components receive power from the Arduino's 5V and GND pins, ensuring stable system operation.

CIRCUIT SET-UP



```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27,16,2);

const int trigPin = 9;
const int echoPin = 10;
const int buzzer = 8;
```

```

long duration;
int distance;

void setup() {

    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    pinMode(buzzer, OUTPUT);

    lcd.init();
    lcd.backlight();

    lcd.setCursor(0,0);
    lcd.print("Parking Sensor");
    delay(2000);
    lcd.clear();
}

void loop() {

    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);

    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);

    digitalWrite(trigPin, LOW);

    duration = pulseIn(echoPin, HIGH);

    distance = duration * 0.034 / 2;

    lcd.setCursor(0,0);
    lcd.print("Distance: ");
    lcd.print(distance);
    lcd.print(" cm ");

    if(distance > 40)
    {
        lcd.setCursor(0,1);
        lcd.print("Safe Distance ");
        noTone(buzzer);
    }

    else if(distance > 20)
    {
        lcd.setCursor(0,1);
        lcd.print("Getting Close ");
        tone(buzzer,1000);
        delay(500);
        noTone(buzzer);
    }

    else if(distance > 10)

```

```

{
  lcd.setCursor(0,1);
  lcd.print("Very Close  ");
  tone(buzzer,1000);
  delay(200);
  noTone(buzzer);
}

else
{
  lcd.setCursor(0,1);
  lcd.print("STOP!    ");
  tone(buzzer,1500);
}

delay(200);
}

```

MANUAL USER GUIDE

1. How to Power ON the System

To power the Parking Assist System, connect the Arduino Uno to a computer or a 5V power supply using a USB cable. Once powered on, the LCD screen will display:

“Parking Assist”

This indicates that the system is ready to start measuring distance.

2. How Input is Given

The ultrasonic sensor automatically detects the distance between the system and an object in front of it.

The sensor continuously sends ultrasonic signals and measures how long it takes for the echo to return. The Arduino then calculates the distance and updates the LCD display in real time.

No manual input is required because the system works automatically.

3. How the System Responds

The system responds differently depending on the detected distance:

Safe Distance (>40 cm)

- LCD: "Safe Distance"
- Buzzer: OFF

Getting Close (20–40 cm)

- LCD: "Getting Close"
- Buzzer: Slow Beeping

Very Close (10–20 cm)

- LCD: "Very Close"
- Buzzer: Faster Beeping

Stop Distance (<10 cm)

- LCD: "STOP!"
- Buzzer: Continuous Sound

4. Meaning of Outputs

LCD Display

Shows the parking status and distance measurement.

Buzzer

Provides sound alerts to warn the driver about nearby obstacles.

Slow beeps indicate that the object is approaching, while fast or continuous sounds indicate that the object is very close.

5. Safety Precautions

- Ensure all wires are properly connected before powering the system.
- Use only a 5V power supply for the Arduino.

- Avoid touching exposed wires while the system is powered.
- Do not place objects too close to the ultrasonic sensor abruptly.
- Keep the circuit on a stable surface to avoid disconnection.

USEFULNESS OF THE SYSTEM

The Parking Assist System helps drivers park vehicles more safely by providing real-time information about the distance between the vehicle and nearby objects. Many minor accidents occur during parking due to limited visibility or incorrect distance estimation.

By using ultrasonic sensing technology, the system can detect objects early and warn the driver through visual and audio signals. This reduces the risk of collisions with walls, other vehicles, or obstacles.

The system operates automatically and continuously monitors the distance in front of the vehicle. Because of this, drivers receive immediate feedback when the vehicle gets too close to an object.

REAL-LIFE APPLICATION

The Parking Assist System can be used in various real-life situations such as:

- Cars and vehicles with parking assistance
- Smart parking systems
- Driver safety training systems
- Robotics and obstacle detection projects

WHO CAN BENEFIT FROM IT

- Drivers who need assistance when parking in tight spaces
- Car manufacturers developing vehicle safety features
- Driving schools teaching safe parking techniques
- Engineers and students studying embedded systems

HOW IT IMPROVES SAFETY AND CONVENIENCE

Efficiency

The system continuously measures distance and provides immediate feedback to the driver.

Safety

It helps prevent minor accidents and damage during parking.

Automation

The system works automatically without requiring manual control.

Convenience

Drivers can easily see the distance and hear warnings while parking.

POSSIBLE FUTURE IMPROVEMENTS

Future improvements of the Parking Assist System may include:

- Adding multiple ultrasonic sensors for wider detection coverage
- Integrating LED indicators to provide additional visual alerts
- Adding wireless connectivity for mobile monitoring
- Implementing a camera system for visual parking guidance
- Integrating the system with a mobile application

These improvements would make the system more advanced and closer to the parking assistance systems used in modern vehicles.

